J!

Royal Rumble

DESIGN AND RULES

A thirty-player elimination trivia format built on the Jeopardy! clue structure and the entry mechanic of the 1980s Royal Rumble. This document covers the rules in full, the optional mechanics, and the design work underneath — including the two problems that nearly broke the format and what the numbers said about fixing them.

Every figure here comes from the simulator or from a recorded match. None of it is illustrative.

THE SHAPE OF IT

Three players start. The rest wait in draw order and enter one at a time as the match runs. Answer a clue correctly and every other player in the ring pays you its value. Miss and you pay, and the clue goes back out to everyone else. Fall below zero and you are gone. The last player standing wins, and nothing else counts as a result.

A full field takes about an hour. The board is six Jeopardy! categories at a time, drawn from an archive of 47,000 categories or from boards written in house, in whatever ratio the host sets.

> ### The one number that matters
>
> Under the obvious scoring rule — a correct answer is worth the clue value, the way it works on television — the player who drew the last entry number won **100%** of simulated matches. Not most. All of them. Everything in the section on scoring exists because of that number.

THE RULES

The board

Six categories are live at once, five clues each, $100 to $500 by row. Row position sets the value regardless of where the category came from — a category lifted from a Double Jeopardy round plays at the same values as any other, and is simply harder. When a category is used up it is replaced immediately, so the board never runs out.

The host can reroll any category before or during play. Three rerolls of the same category retires it permanently.

Entering

Every player draws an entry number before the match. Numbers one to three open the match; the rest enter on a fixed clue interval, which the host sets or lets the app scale to the size of the field. Each entrant walks in with the same stake.

Scoring

Correct. Every other player in the ring pays you the clue value. You collect all of it. With five in the ring a $500 clue is $2,000 to you and $500 from each of the other four; heads-up it is worth $500.

Wrong. You lose the value and you are locked out of that clue. Nobody else is affected by your miss.

The re-toss. A missed clue immediately goes back out to everyone still eligible as a fresh buzzer race. If somebody else converts it you pay them along with everyone else, so a miss followed by a conversion costs you twice.

Nobody gets it. Every player in the ring loses half the value. Passing is not free.

Elimination

Below zero and you are out; exactly zero survives. If one answer sinks several players they all go together, and the player whose answer did it is credited with a **toss out** for each. If a clue would wipe the entire ring, the highest score survives.

The ceiling

There is a maximum score, and it falls across the match. Anything above it is clipped. It never drops below the amount a new entrant carries in, so nobody arrives already capped.

Clearing the field

Eliminate everyone in the ring while players are still queued and you take a bonus equal to every clue left on the board. The board is replaced entirely and two fresh players enter rather than one.

Buzzing

The host reads the clue, then arms the buzzers. Fastest reaction takes it, one buzz per player per race. Buzzing before the lights costs a 250 millisecond lockout, and the penalty runs even if the buzzers open while it is being served.

Players still in the queue, and players already eliminated, can buzz on every clue. It does not score and cannot affect the match, but it reports their time and where they would have placed against the live field. Arriving with your timing already calibrated is a real advantage.

> **Times under 150 milliseconds are normal**
>
> Strong players do not react to the lights. They learn the rhythm of the host's read and time the buzzer to land the instant it opens. A perfectly judged buzz reads 0.0. In the first recorded match, the two active players buzzed a median of 197 and 207 milliseconds with best times of 8 and 17 — and one of them jumped the lights on a third of his presses. That is the trade the lockout exists to price.

WHY THE SCORING WORKS THIS WAY

The draw-position problem

The natural rule is the televised one: a correct answer is worth the clue value to the player who gets it, and the format supplies the drain by taking that value off everyone else. Simulated across four thousand thirty-player matches, that rule produced this:

Draw numbers	Share of wins	Fair share
1 – 10	0.0%	33.3%
11 – 20	0.0%	33.3%
21 – 25	3.1%	16.7%
26 – 30	96.9%	16.7%
— of which draw 30 alone	46.7%	3.3%

Four thousand simulated matches. Draws 1 through 21 did not win once.

The cause is arithmetic rather than luck. Under flat scoring a player’s expected change per clue is $V(2 – P)/P$, where V is the clue value and P the number of players in the ring. For any P above two that is negative for everybody. Nobody can accumulate; the whole field drifts downward at roughly the same rate. Across an entire thirty-player match the highest score anyone reached was around 5,950 against a 5,000 starting stake — so the score ceiling never bound, the field-clear bonus almost never triggered, and a fresh entrant arriving on 5,000 simply outlasted incumbents who had been grinding down for half an hour.

The fix: collect from each opponent

The loss side is left alone — it is what makes the format an elimination game. The gain side changes: the answerer collects the clue value from *every* opponent rather than once. Elimination pace is untouched, but a strong player can now build a wall, which is what allows an early entrant to survive long enough to face the last numbers.

It is also the more natural sentence to say out loud at the table: *everyone else pays you.*

The ceiling, and why it falls

Letting players accumulate reintroduces a different problem: a runaway leader. A fixed ceiling solves that but hands the advantage straight back to the late draws, because it clamps exactly the players who have been building. A *rising* ceiling is worse still — it is tight while the early entrants need room and generous by the time the last numbers arrive.

Ceiling	Match length	Back-half draws win
Fixed 6,000	67 min	72%
Fixed 10,000	118 min	55%
Rising, 4,000 +40/clue	100 min	73%
Falling, 15,000 –50/clue	76 min	53%

A falling ceiling reaches near-perfect draw fairness at 76 minutes, where a fixed one needs 118 minutes to get there. 50% is fair.

So the ceiling decays. Early entrants get room to build; by the time the last numbers walk in, the ceiling has compressed and there is no worn-down field to exploit. The floor is the entry stake, so a late arrival is never capped on the way in.

What the tuning produced

Field	Entry every	Start	Ceiling	Decay	Length	Back-half
10	10 clues	3,000	7,500	–40	35 min	57%
16	7 clues	3,000	7,500	–25	46 min	62%
20	6 clues	3,000	7,500	–25	48 min	64%
30	5 clues	3,000	11,000	–40	60 min	63%

Back-half is the win rate of players drawing in the second half of the field. It started at 100%.

A late number is still worth having, and that is deliberate — it is true to the source material, where entrants 27 through 30 genuinely do overperform. What it is no longer is a guarantee.

OVERTIME

Two evenly matched players trade the same points back and forth indefinitely. Heads-up, a correct answer is worth exactly what the opponent loses, so a field of two with similar ability is a random walk with no drift. It showed up in the first recorded match: the last fourteen clues oscillated without either player getting meaningfully closer to going out.

Once the queue is empty and the ring is down to two, clue values double every six clues, capped at eight times face. The board displays the raised values, and the room gets a full-screen announcement at each step.

Endgame	Result
Perfectly alternating stall, no overtime	Still running at 400 clues

Perfectly alternating stall, with overtime	Resolved in 22 clues

The pathological case, since real players are never quite this symmetrical.

OPTIONAL MECHANICS

Four rules that change how the game is played rather than how it looks. All off by default, each toggled separately, and all of them lean on keyboard controls — worth leaving off if the field is mostly on phones.

Top rope

Declared between clues, never once a clue is on the board — otherwise you would only ever climb up when you already knew the answer. That clue is worth double in both directions to the player who declared, and their winnings ignore the ceiling. That last part matters: without it the top rope is strictly bad for anyone near the cap, since the downside is uncapped and the upside is not.

Targeting

Aim at one player, visible to the room, with an alert on their buzzer. Win the clue and the entire pot comes out of them alone and everyone else is spared. Lose the clue to them and you pay the whole pot yourself while the rest of the ring walks. A finishing move and a kamikaze on the same button.

Bounties

A player waiting in the queue stakes part of their own entry on a head, up to half. Whoever eliminates the target collects. If the target survives to the end they keep it — and if the target eliminates the player who placed it, they keep that too. Placing a bounty is a declaration, not a free shot.

Revival

An eliminated player returns to the queue at a fraction of the stake with a new entry number, once by default.

Setting	Length	Back-half draws win
None (baseline)	64 min	53%
Top rope	62 min	54%
Targeting	63 min	63%
Bounties	64 min	48%
Revival	95 min	52%
All four	88 min	63%

400 simulated thirty-player matches per row.

Three of the four barely move the clock. Revival runs half again as long, because nearly every player spends their second life — the setup screen says so before the match starts. It is also the fairest setting on the list, which was not the expectation: a second chance is worth most to whoever went in first, so it works against the same late-draw advantage the falling ceiling exists to fight.

> ### A measurement trap worth recording
>
> Revival first showed a 94% back-half win rate, which looked alarming. It was an artefact: revived players receive a fresh entry number, so the winner is a late draw almost by definition. Measured against the number each player *actually drew*, it is 52%. Players now carry both numbers and the standings report the drawn one.

RUNNING A MATCH

Before

The setup screen mints a four-letter room code to read aloud, and fills a lobby as players join. Clue material is mixed by percentage across the archive, boards written in house, and anything uploaded for that match; the archive can be narrowed by season. Fresh boards can be dropped in as j-trivia.org JSON or jparty.tv CSV.

The screen estimates the length from the roster and warns when the numbers do not work — a six-player field cannot be stretched to an hour by entry pacing alone, and it says so, with a button that sets a target it can actually reach.

During

- The clue takes over the shared screen. Correct responses appear only in a separate admin window, kept on an unshared display.
- Three keys resolve everything: correct, wrong with automatic re-toss, and nobody got it.
- A five-second lectern countdown runs after the buzzers arm.
- Undo takes back an entire clue — scores, eliminations, entries, the board. Clicking any player adjusts their score directly. Both are logged.
- The audio delay is adjustable mid-match, because nobody can tell you it is wrong until they have tried to buzz on a real clue.

Timing

The sockets are much faster than the call audio, so buzzers are deliberately held back to arm when the host's voice actually arrives. Measured on a real match with players spread across the country:

Path	One way
Socket, fastest player	7 ms
Socket, median player	14 ms
Socket, slowest player	29 ms
Zoom audio, typical	150 – 300 ms

Socket figures from a recorded match; the spread within each player was only a few milliseconds. Zoom is the industry-typical range.

So the socket beats the voice by roughly 120 to 290 milliseconds, and that gap is what the delay setting closes. Players also get their own trim in ten millisecond steps, since audio paths differ by client, buffer, and whether somebody is on headphones.

After

The winner gets the screen. Everything else — clues survived, toss outs, points drained, peak score, buzzer attempts, win rate, average and best times, the single fastest buzz of the match — goes into sortable standings and a score graph, shareable as a summary card, a full stats sheet, or one wide CSV.

With recording switched on, the match also produces a detailed log: every clue, every buzz time and the connection it arrived over, scores before and after, entries, eliminations, corrections, and a comparison of the predicted length against the real one. That file is how the model gets better.

> **What a recording changed**
>
> The first recorded match produced a median pace of 15.0 seconds per clue against a mean of 21.2 — six discussion breaks between 30 and 57 seconds dragged the average. It also exposed two broken statistics and an endgame that could not end. Three rules and four bugs came out of one file.

METHOD

The rules engine is a plain module with no framework and no network, driven by a Monte Carlo harness that plays hundreds of matches per configuration using a simple model of player ability. Every figure in this document comes from that harness or from a recorded match.

The harness runs before every deployment. It has already caught a scoring change that pushed non-pot matches from 57 minutes to 746 — a regression that was invisible in ordinary play and would have been found on a Saturday night otherwise.

Clue material comes from a public dataset rather than from j-archive directly. The archive’s maintainer asked not to be crawled, and a slow crawler is still a crawler.